

① $V - V - F - V$

② a) 0×562

0101 1100 0010 0000 0101 1100 00102

b) $2048 \cdot 10^6 = 2.048 \text{ 000 000 bytes}$

$\frac{2.048 \text{ 000 000}}{1.073 \text{ 741 824}} = 1.907 \text{ G.B}$

c) 10101010

$\begin{array}{r} 10101010 \\ - 01001101 \\ \hline 01100111 \end{array}$

③ a) -57_{10}

57 $\frac{128}{64} \quad \frac{32}{16} \quad \frac{8}{4} \quad \frac{2}{1}$

$\begin{array}{r} 57 \\ - 32 \\ \hline 25 \\ - 16 \\ \hline 9 \\ - 8 \\ \hline 1 \end{array}$

00111001 $C1 = 11000110$

b) 10011010_{10}

$C1: 01100101$

$\begin{array}{l} 1 \cdot 2^0 = 1 \\ 0 \cdot 2^1 = 0 \\ 1 \cdot 2^2 = 4 \\ 0 \cdot 2^3 = 0 \\ 0 \cdot 2^4 = 0 \\ 1 \cdot 2^5 = 32 \\ 1 \cdot 2^6 = 64 \\ 0 \cdot 2^7 = 0 \end{array}$

$\begin{array}{r} 64 \\ + 32 \\ \hline 96 \end{array}$

$R: -101_{10}$

c) 01010000

$\begin{array}{r} 01010000 \\ + 01110000 \\ \hline 11000000 \end{array}$

→ overflow

$$④ \text{ a) } CPI_g = \frac{10^6 \cdot (400 \cdot 1 + 100 \cdot 3 + 150 \cdot 4 + 50 \cdot 4)}{10^6 \cdot (400 + 100 + 150 + 50)} = \frac{1500}{2500} \approx 2,142$$

$$⑤ \text{ b) } T_{cpu} = \frac{CPI \cdot I}{F \cdot I} = \frac{1500 \cdot 10^6}{2,5 \cdot 10^9} = \frac{1500 \cdot 10^6}{2500 \cdot 10^6} = 0,6 \mu s$$

$$⑥ \text{ c) } CPI = \frac{10^6 \cdot (400 \cdot 1 + 100 \cdot 3 + 75 \cdot 4 + 50 \cdot 4)}{10^6 \cdot (400 + 100 + 75 + 50)} = \frac{1700}{625} = 1,92$$

$$T = \frac{1200 \cdot 10^6}{2500 \cdot 10^9} = 0,48 \mu s$$

$$Speedup = \frac{0,6}{0,48} = 1,25 \times$$

⑦

with $i=0$;

while ($i < n$) {

$C[i] = A[i] + K$;

$i++$;

}

⑧ limits:

addi $\$sp, \$sp, -8$

sw $\$ra, 0(\$sp)$

addi $\$to, \$a0, \$a2$

sw $\$a0, 4(\$sp)$

add $\$a0, \$zero, \$to$

jal mure

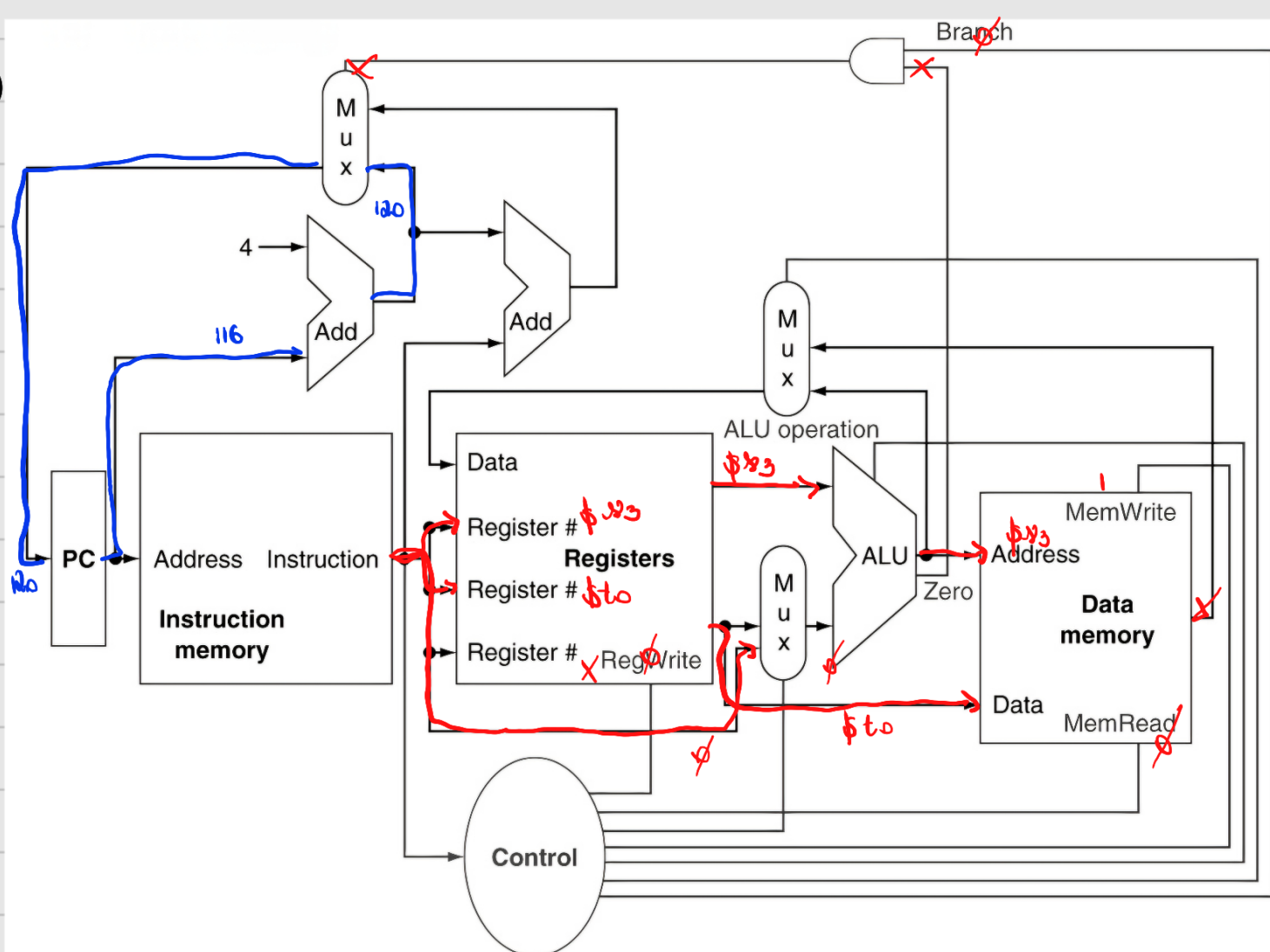
sub $\$v0, \$v0, \$a2$

sw $\$ra, 0(\$sp)$

sw $\$a0, 4(\$sp)$

jr $\$ra$

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instrução: sw \$t0, 20(\$s3)

entradas: $\$s3 = 100$; $\$t0 = 58$

PC: 116